Auctions & Mechanism Design Basics

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- We study about a kind of science of *rule-making*.
- To make it simple, we first consider single-item auctions.
- We will go over some basics about first-price auctions and second-price auctions.
- Also, we will talk about
  - incentive guarantees,
  - strong performance guarantees, and
  - computational efficiency

  in an auction.
- We will end the discussion with Myerson's Lemma.



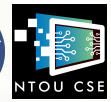

Outline

1. Single-Item Auctions
2. Sealed-Bid Auctions
   - First-Price Auctions
   - Second-Price Auctions
   - Case Study: Sponsored Search Auctions



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- Consider a seller with a single item.
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  - v_i is private.
    - Unknown to the seller and other bidders.



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(i) Each bidder i privately communicates a bid b_i to the seller—in a sealed envelope.

(ii) The seller decides who gets the item (if any).

(iii) The seller decides the selling price.



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- Step (ii): The selection rule. We consider giving the item to the **highest** bidder.



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The winning bidder pays her bid.

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Issues of the First-Price Auctions

- For a bidder:



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Issues of the First-Price Auctions

- For a bidder: Hard to figure how to bid.
- For the seller: Hard to predict what will happen.



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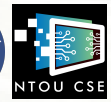
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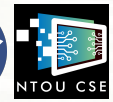
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  - Would your answer change if you knew there were two other bidders rather than one?

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    - Your maximum bid is reached, or
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    whichever comes first.
  - For example, if the highest other bid is \$90.
    You only pay \$90 $+ \epsilon$ for some small increment ϵ.

  $\approx$ highest other bid!



Second-Price auction

Second-Price/Vickrey Auction

The highest bidder wins and pays a price equal to the second-highest bid.

- Is such a strategy a dominant strategy?



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- Is such a strategy a dominant strategy?
  - The strategy is guaranteed to **maximize** a bidder's utility **no matter what other bidders do**.



Truthfully Bidding Is Dominant Here

Proposition (Incentives in Second-Price Auctions)

In a second-price auction, every bidder i has a dominant strategy: set the bid $b_i = v_i$, equal to her private valuation.



Proof of the Proposition

- Fix a bidder i with valuation v_i.
- $\boldsymbol{b}$: the vector of all bids.
- $\boldsymbol{b}_{-i}$: the vector of $\boldsymbol{b}$ with b_i removed.

* **Goal**: Show that bidder i's utility is maximized by setting $b_i = v_i$.



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  - $\because$ bidder i wins and bids her true valuation v_i, so $p \leq v_i \Rightarrow v_i - p \geq 0$.



Second-Price Single-Item Auctions are "ideal"

Definition (Dominant-Strategy Incentive Compatible)

An auction is dominant-strategy incentive compatible (DSIC) if

- truthful bidding is a dominant strategy for every bidder, and
- truthful bidders always obtain nonnegative utility.



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where $\sum_{i=1}^{n} x_i \leq 1$; $x_i = 1$ if bidder i wins and 0 if she loses.

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- So such an auction is welfare maximizing if bids are truthful.



Second-Price Single-Item Auctions are "ideal" (contd.)

Theorem

A second-price single-item auction satisfies:

(1) DSIC.

(2) Welfare maximizing.

(3) It can be implemented in polynomial time.

In fact, (3) is linear.



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A second-price single-item auction satisfies:

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How a sponsored search auction works
Advertiser pays for the click
Camera Store A
Advertisers submit bids
Bid
Camera Store B
Bid
Search engine
(auction)
Auction selects and orders ads
Camera Store C
Bid
User searches for camera
Search
https://www.search.com
Search
camera
All
Images
Shopping
Videos
News
More
Sponsored
Ad slot 1
Camera Store A
Wide selection of cameras for every creator. Shop DSLRs, mirrorless and more.
Click
Opens the advertiser's website
Sponsored
Ad slot 2
Camera Store B
Quality cameras and lenses for photo and video. Great selection and expert support.
Camera buying guide
How to choose the right camera for your needs, with key features, types and tips.
Camera reviews
In-depth reviews of popular cameras, with hands-on tests and comparisons.
Camera Store A
https://www.camerastorea.com
Camera Store A
Cameras
Lenses
Accessories
Find the right camera for your creative journey
Explore cameras
The user visits the shop; the advertiser pays the search engine.

Sponsored search: from a query to an outcome

3 advertisers / 2 ad slots / Myerson (VCG) payment rule / No reserve

1 A user searches camera

Public inputs
Slot CTRs: $\alpha_1 = 0.40$, $\alpha_2 = 0.20$; all quality scores $\beta_i = 1$.

2 Advertisers bid

One bid per click, for this query.

Advertiser	Private value v_i	Bid b_i
A	$10	$10
B	$6	$6
C	$2	$2

The platform receives the bids b_i.
Here, all bid truthfully: $b_i = v_i$.

3 Assign the slots

Rank bids: $10 > 6 > 2$.

Results for: camera

- **A** Advertiser A — Ad — Slot 1 | CTR = 0.40
- **B** Advertiser B — Ad — Slot 2 | CTR = 0.20

C gets no slot.

$x = (x_A, x_B, x_C) = (0.40, 0.20, 0)$

4 Click & charge

Prices are set before clicks occur.

- **A** Click on A — A pays $4
- **B** Click on B — B pays $2

No click on an ad: no charge for that ad.

Expected payments per page view:
$p_A = 6(0.4 - 0.2) + 2(0.2) = 1.60$
$p_B = 2(0.2) = 0.40$
Price per click $= p_i/x_i$ for each winner.

5 Expected outcome per page impression

One impression = one results-page display

Advertiser	Slot	Assigned CTR x_i	Expected value $v_i x_i$	Expected payment p_i	Expected utility $v_i x_i - p_i$
A	1	**0.40**	$4.00	$1.60	$2.40
B	2	**0.20**	$1.20	$0.40	$0.80
C	None	**0**	$0	$0	$0

Social welfare: $4.00 + 1.20 = \$5.20$ **Seller revenue:** $1.60 + 0.40 = \$2.00$

WHY SINGLE-PARAMETER? One private value per click v_i determines every slot value $v_i\alpha_j$.

Background

The Social Dilemma (2020) - Trailer

- Web search results:
  - relevant to your query (by an algorithm, e.g., PageRank).
  - pops out a list of sponsored links.
    - They are paid by advertisers.
- Every time you give a search query into a search engine, an auction is run in real time to decide
  - which advertiser's links are shown,
  - how these links are arranged visually,
  - what the advertisers are charged.




Multiple Items for Sponsored Search Auctions

- Let's say the items for sale are k "slots" on a search results page.
- Bidders: the advertisers who have a bid on the keyword that was searched on.
  - On the keyword, "university", NTU, NYCU, NCKU, NTOU, etc., might be the bidders.



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  - On the keyword, "camera", Nikon, Canon, Sony, etc., might be the bidders.
  - On the keyword, "SUV", Toyota, Ford, Honda, Porsche, etc., might be the bidders.
- Let's say the items are not identical.
  - Higher slots are more valuable. What do you think?



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  - The probability that the user clicks on this slot.
  - Assumption: $\alpha_1 \geq \alpha_2 \geq \ldots \geq \alpha_k$.



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  - Here we simply assume that $\beta_i = 1$ for each i.
- The expected value derived by advertiser i from slot j: $v_i \alpha_j$
- The social welfare is $\sum_{i=1}^{n} v_i x_i$.
  - x_i: the CTR of the slot to which bidder i is assigned.
    - $x_i = 0$: bidder i is not assigned to a slot.
  - Each slot can only be assigned to one bidder; each bidder gets (at most) one slot.



Our Design Approach

- Who wins what?
- Who pays what?
- The payment.



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- Who wins what?
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- The payment.
  - If the payments are not just right, then the strategic bidders will game the system.




Our Design Approach

Design Steps

(a): Assume that the bidders bid truthfully. Then, how should we assign bidders to slots so that property (2) and (3) holds?

(b): Given the answer of above, how should we set selling prices so that property (1) holds?



Step (a)

- Given truthful bids. For $i = 1, 2, \ldots, k$, assign the ith highest bid to the ith best slot.



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- You can prove that this assignment achieves the maximum social welfare as an exercise.



Step (b)

- There is an analog of the second-price rule.
  - DSIC.
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  - DSIC.
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  - Why not just ask the bidders to pay the bid which is next (lower) to its own?



Step (b)

- There is an analog of the second-price rule.
  - DSIC.
  - ⋆ Myerson's lemma.
    - A powerful and general tool for implementing this second step.
  - Why not just ask the bidders to pay the bid which is next (lower) to its own?
    - ▷ Generalized Second Price Auction.

